Auditing Data Leakage and Reporting Gaps in EEG Connectivity Deep Learning for Depression Detection

Kunal Chevuri
Prosper High School
Prosper, TX, USA
kunalchevuri510@gmail.com

Leah Ileto
St. Lucy's Priory High School
Glendora, CA, USA
leahileto07@gmail.com

Mohammed Aledhari
University of North Texas
Denton, TX, USA
Mohammed.Aledhari@unt.edu

***Abstract*—Deep learning applied to electroencephalographic (EEG) functional connectivity is a prominent approach to detecting major depressive disorder (MDD), with reported accuracies often above 90%. Recent reviews have examined how these studies split data into training and test sets, but not the connectivity pipeline that produces the model input: connectivity estimation, thresholding, normalization, and channel or band selection performed before partitioning can transfer test information into training, and epoch-level partitioning can place segments from one subject on both sides of the split. We define five such leakage surfaces and a four-level risk rating based on reported methods alone, and apply the rating to 18 studies from a citation-ranked corpus rather than a systematic search, so the figures below describe this corpus and are not prevalence estimates for the literature. The clearest finding is partitioning: 50% of the audited studies partitioned at the epoch rather than the subject level, the surface on which two independent raters agreed most closely ($\kappa = 0.91$). A further 22% left at least one preprocessing step undescribed. No study reported fitting connectivity estimation, thresholds, or normalization on the whole dataset; beyond partitioning, the detectable risk came from selecting channels or bands on all data (17% of studies). Because a step a paper never mentions cannot be distinguished from one never performed, we also report the strictest reading of that convention, under which 0 studies can be verified leakage-free. Only 2 studies rated leakage-free reported an eligible accuracy, too few to test whether accuracy differs by rating. We propose an eight-item reporting checklist that lets a reader verify, from the paper alone, that a connectivity pipeline is free of leakage.**

***Index Terms*—electroencephalography, functional connectivity, deep learning, major depressive disorder, data leakage**

I. Introduction

A. Clinical motivation

Major depressive disorder (MDD) is among the leading causes of disability worldwide [1], yet its diagnosis still rests on clinical interviews and self-report scales of limited test-retest reliability [2], [3]. An objective physiological marker would help, and electroencephalography (EEG) is an attractive candidate: affordable, non-invasive, portable, and resolved on the millisecond scale [4]. Depression is increasingly understood to involve abnormal communication between large-scale brain networks [5], and functional connectivity—the statistical dependence between signals at different sites, estimated with measures such as coherence, the phase-locking value or the phase lag index—captures that communication from scalp EEG. Deep networks learn from connectivity matrices directly, and a substantial literature applies them to separating people with MDD from healthy controls, many reporting accuracies above 90% [6].

B. The verification problem

Accuracies of this magnitude would be clinically meaningful if they held. Whether they hold depends on something a reader usually cannot check: whether any information from the test data reached the model during training. This is data leakage, and in neuroimaging machine learning it has been shown to inflate reported performance substantially [7]–[9].

Recent reviews of EEG-based depression detection have examined how studies validate their models—holdout and cross-validation schemes, per-subject and leave-one-subject-out partitioning, and whether models were tested on independent datasets [2], [10]. That is the right question about the split. But a connectivity model does not consume raw EEG. It consumes a connectivity matrix, built through a chain of steps that each run before the split: recordings are epoched, connectivity is estimated, matrices are thresholded into graphs, values are normalized, and channels or frequency bands are selected. Each step can carry test information into training if it draws on data beyond the training fold, and most papers do not report the order of these steps, which leaves the question unanswerable from the paper alone. Fig. 1 contrasts a leakage-free ordering with a leaking one and marks the five points (L1–L5) at which test information can enter.

Prior reviews audit how these studies split their data. This work audits what happens to the data before it is split. We searched PubMed, OpenAlex, Semantic Scholar and Google Scholar through September 2026 and found no review that audits connectivity-pipeline construction as a leakage surface in this literature; because that search was not itself conducted as a systematic review, we report it as the limit of our own searching rather than as a claim of priority.

We ask four questions. How completely do studies report each leakage surface (RQ1)? Can leakage risk be rated reli-

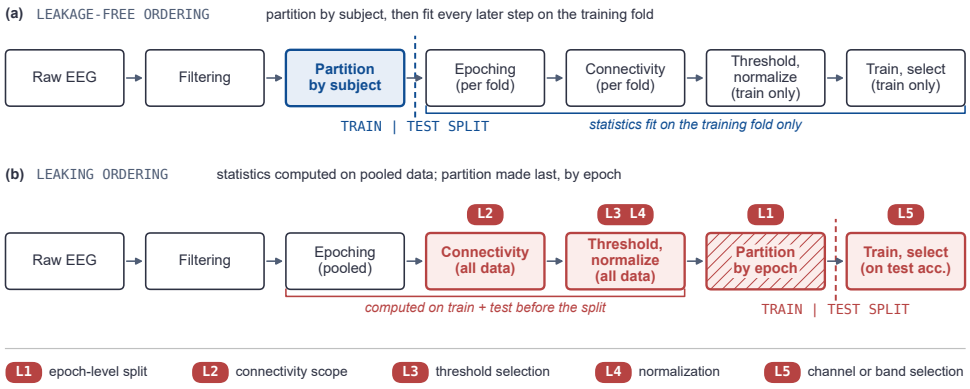


Fig. 1. The connectivity pipeline with leakage surfaces marked. (a) Leakage-free ordering: data are partitioned by subject before any statistic is computed, and every later step uses the training fold only. (b) Leaking ordering: connectivity (L2) and threshold or normalization statistics (L3, L4) are computed on pooled data, the partition is made by epoch so one subject can fall on both sides (L1), and channels or bands are chosen by test-set accuracy (L5). The dashed line marks the split. Red marks leakage points, hatching the epoch-level partition.

ably from reported methods by independent raters (RQ2)? Is reported accuracy associated with the rating (RQ3)? And, as a secondary question, does reporting completeness change with publication year (RQ4)?

C. Contributions

1) A leakage taxonomy for EEG connectivity pipelines. We identify five points at which construction of a connectivity matrix can allow test information into training, and describe the conditions under which each occurs.
2) A leakage-risk rating scheme. We define a four-level rating applied from reported methods alone, with explicit decision rules, and report inter-rater agreement.
3) An audit of the literature. We apply the scheme to 18 studies and report how completely pipelines are described and how reported accuracy varies across risk strata.
4) A reporting checklist. We propose a short set of items that connectivity-based studies should state so leakage can be ruled out by a reader.

II. Related Work

A. Deep learning on EEG connectivity for MDD

In this literature a connectivity measure turns each EEG segment into a channel-by-channel matrix, and a deep network classifies the matrices. Most measures are undirected phase or amplitude couplings (PLV, PLI, coherence, correlation and mutual information) [11]–[15]; a smaller group uses directed measures from multivariate autoregressive models [16], [17]. Some stack one matrix per frequency band into a multilayer network [12], [18]. Convolutional networks treat the matrix as an image, sometimes with a recurrent branch [13], [16], [19]; graph neural networks treat channels as nodes and connectivity as edge weights [15], [18], [20], and recent work learns the graph during training [21], [22] or moves it to cortical source space [23]. A few public datasets with tens of participants per group recur (Table I). Reported accuracies commonly exceed 95%, yet these studies seldom state when the pipeline's statistics were computed relative to the train/test split.

B. Existing reviews of validation practice

Recent reviews compare validation schemes in EEG depression studies; one found none that validated across datasets [2], another found near-perfect accuracy most often where samples were small and validation subject-dependent [10], and a third found that the lack of standardized evaluation protocols limits comparison [6]. A meta-analysis of 176 machine-learning studies of MDD found reported accuracy negatively correlated with sample size [24]—the confound this audit addresses through sample-size stratification.

The distinction we draw is one of scope. These reviews ask how a study separated training from test data, and whether its validation was subject-dependent or subject-independent, and one of them also discusses feature selection [10]. None that we found asks whether the steps that build the connectivity matrix itself—estimation, thresholding and normalization—were fit before that separation. A study can use leave-one-subject-out validation and still leak through a threshold chosen on the pooled data; a review that records only the validation scheme would score it as sound.

C. Data leakage in neuroimaging machine learning

Leakage is well documented in neuroimaging. Across four resting-state fMRI connectome datasets, selecting features on the full dataset and repeating subjects across the split inflated performance most [7]. In MRI classification, splitting by slice rather than subject raised a VGG16 model's accuracy on one dataset from 66% to about 97% [8]. In resting-state EEG from patients with post-traumatic stress disorder, splitting cropped trials rather than participants inflated balanced accuracy by about 5–15 percentage points for one convolutional network and 15–20 for EEGNet [9]. Segment-based rather than subject-based holdout strongly overestimated accuracy on unseen subjects, and most translational EEG studies surveyed used

it [25]; in EEG-based schizophrenia detection, about 65% of studies split by epoch or selected features before partitioning, inflating accuracy by up to about 30% [26]. None examined how connectivity features are built. The connectome work is the closest analogue, deriving features from pairwise connectivity; the two forms it found most damaging correspond to L5 and L1 here. Rating leakage from reported methods also has precedents: one survey labeled each Alzheimer's study's leakage status from its text [27], and a scoping review stratified 44 such studies into three risk tiers, finding reported accuracy highest where risk was highest [28]. We adapt that approach to the steps that build a connectivity matrix.

III. Leakage in Connectivity Pipelines

A. Pipeline formalization and connectivity measures

Let $x_i(t)$ denote the signal recorded at channel i, $i = 1, \ldots, N$, and let T be the number of samples in an epoch. A connectivity-based classifier does not operate on $x_i(t)$ directly; it operates on a matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ derived from $x_i(t)$ through a chain of steps—filtering, epoching, connectivity estimation, thresholding, normalization—each of which is fit on some subset of the data. The question this audit asks of each study is which subset that was.

Magnitude-squared coherence between channels i and j at frequency f is

$$C_{ij}(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f)\, S_{jj}(f)}, \tag{1}$$

where $S_{ij}(f)$ is the cross-spectral density [29]. S_{ij} is estimated over a window of data; if that window spans samples from both partitions, or is averaged across subjects before the partition is applied, the estimate and every feature derived from it carry information from outside the training fold.

The *phase-locking value* [30], with $\Delta\phi_{ij}(t)$ the instantaneous phase difference between channels i and j, is

$$\mathrm{PLV}_{ij} = \frac{1}{T}\left|\sum_{t=1}^{T} e^{\mathrm{j}\,\Delta\phi_{ij}(t)}\right|. \tag{2}$$

PLV is bounded in $[0, 1]$ and is usually estimated within a single epoch, which limits but does not remove the estimation route: pooling epochs across subjects before computing it still leaks.

The *phase lag index*, $\mathrm{PLI}_{ij} = \left|\left\langle \mathrm{sign}\left(\Delta\phi_{ij}(t)\right)\right\rangle_t\right|$, discounts zero-lag coupling, which volume conduction would otherwise dominate [31].

Any of these yields a weighted adjacency matrix $\mathbf{A}$. Matrices are often binarized at a threshold τ ($A^{\mathrm{bin}}_{ij} = 1$ if $A_{ij} > \tau$, else 0), or reduced to their k strongest edges. If τ or k is chosen from the distribution of A_{ij} pooled across the full dataset—a common way to fix a target graph density—it is a parameter fit on data that include the test set.

B. A taxonomy of leakage surfaces

Fig. 1 shows two orderings of the same pipeline. In the leakage-free ordering (a), the train/test partition is applied immediately after filtering, and every later step is computed separately within each fold. In the leaking ordering (b), the partition is applied last, after epoching, connectivity estimation, and threshold or normalization statistics have been computed on the pooled dataset, so the test set has already shaped the features before the model sees a split. We identify five points at which this can occur.

L1 — Epoch-level partitioning. A recording contributes many epochs, each treated as a sample. If the train/test or fold assignment is made per epoch rather than per subject, epochs from the same subject appear on both sides. *Leakage-free condition:* every epoch from a subject is assigned to the same side.

L2 — Connectivity estimation scope. Coherence, PLV, PLI, or another measure is estimated from a window of data. *Leakage-free condition:* each estimate lies entirely within one fold, and no estimate is pooled or averaged across subjects before the partition.

L3 — Threshold selection. A threshold τ or edge count k is applied to the matrix. *Leakage-free condition:* τ or k is fixed a priori or chosen from training-fold statistics only, never from a distribution pooled across the full dataset.

L4 — Feature normalization. Matrices or derived features are standardized or rescaled. *Leakage-free condition:* normalization statistics are computed on the training fold and applied unchanged to the test fold.

L5 — Channel or band selection. A subset of electrodes or frequency bands is chosen for the final model. *Leakage-free condition:* the subset is chosen a priori or from training-fold performance only, never by comparing candidate subsets on test-set accuracy.

Section IV turns these conditions into rating rules, and Table II (Section VI) into a checklist a reader can apply.

Out of scope. Independent component analysis, artifact subspace reconstruction, re-referencing, bad-channel interpolation and normalization of the matrix itself can also be fit on the full dataset before the split; we excluded them as generic preprocessing, and because the corpus rarely describes them well enough to code. L1–L5 and Table II are therefore necessary, not sufficient, for ruling out leakage.

C. Why EEG subject signatures make L1 severe

L1 deserves separate emphasis because EEG carries stable subject-specific signatures: resting-state recordings, including channel coherence, identify individuals across sessions at least two weeks apart [32]. A classifier trained on epochs from a subject who also appears in the test set does not need to learn anything about depression to score well on that subject's test epochs; it can recognize the individual and recall their label. A 2026 preprint reproduced this inflation from shared subjects across five resting-state cohorts, including eyes-closed and eyes-open recordings from the Mumtaz depression dataset [33]. The literature under audit typically pairs tens of subjects (median 61, Table I) with many epochs per subject, which is the regime in which within-subject information dominates a shuffled split. The effect is compounded, not replaced, by

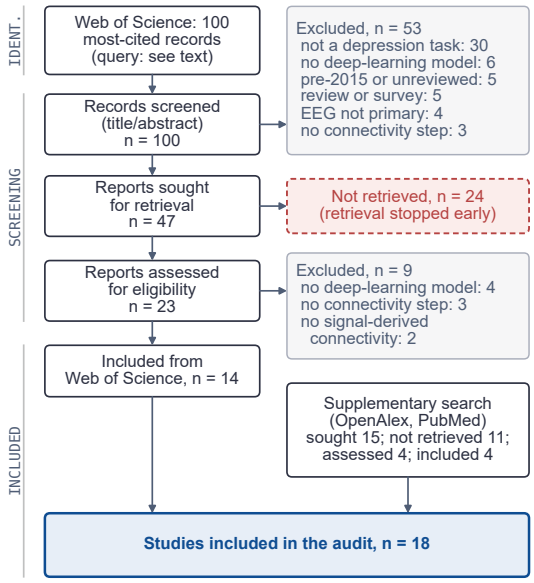


Fig. 2. PRISMA-style screening flow: records at each stage and the reasons for exclusion, generated from the screening log. It is not a systematic review's flow—the starting set was citation-ranked and retrieval incomplete (Section VII). The supplementary search was run when fewer than 15 studies had qualified, and retrieval stopped at 15 (Section IV). Each of the 100 records is counted once. The dashed box marks reports not retrieved.

L2–L5: even under subject-level partitioning, a connectivity or threshold estimate computed on the whole dataset can still transfer test information into the training features.

IV. Methods

A. Search and screening

The starting corpus was a previously assembled set of the 100 most-cited articles returned by a Web of Science search on EEG, connectivity, deep learning and depression. Records were screened on title and abstract, then full text, against six criteria, all of which had to hold: (1) EEG is the primary modality; (2) a functional or effective connectivity measure is computed—coherence, PLV, PLI, correlation, Granger causality, mutual information, or a graph derived from one; (3) a deep learning model is applied to those features; (4) the study classifies depression or predicts its severity; (5) at least one quantitative performance metric is reported; and (6) the work is peer-reviewed, in English, and published in 2015 or later. Criterion 2 required at least one edge weight to depend on the recorded signals, so a graph fixed by electrode positions or an atlas did not qualify however the paper described it. Because fewer than 15 Web of Science studies had qualified, the protocol's supplementary search was run in OpenAlex and PubMed in September 2026; retrieval then stopped once the corpus reached 15, a stopping rule the protocol did not specify, so 24 screened-in Web of Science reports were never assessed at full text, one because no copy was available to us. Fig. 2 gives the counts and exclusion reasons.

B. Extraction fields

The coding sheet's columns were fixed before any study was coded. Descriptive fields recorded the citation; the dataset (MODMA, PRED+CT, the Mumtaz resting-state dataset, a private cohort, or other); total, MDD and control sample sizes; the connectivity measure; the frequency bands; and the architecture family. Pipeline fields recorded whether recordings were epoched and how many epochs each subject contributed; the partitioning unit (subject, epoch, or unclear); and the validation strategy (holdout, k-fold, leave-one-subject-out, external, or unclear). Four further fields recorded, for each of L2–L5, whether the step was stated to be fit within the training fold, fit on the whole dataset, not performed, or unclear.

Performance fields recorded reported accuracy and other metrics. Where a study reported more than one accuracy, we took the result for the model the paper proposes under the most subject-independent protocol it reports, coded its leakage fields for that same protocol, and preferred the results tables where these disagreed with the abstract. This rule was fixed after the initial coding pass, which had followed each abstract's headline figure, but before any accuracy was analyzed; one study was left on the protocol both raters coded, with the matching accuracy, so that rating and accuracy always refer to the same protocol. The abstract's headline figure was kept as a second field for a sensitivity analysis. Only binary depression-versus-control accuracies enter the primary comparison; other tasks enter a sensitivity analysis.

For every study each rater recorded an evidence quote, the passage on which the coding rests. Coding drew only on the published text: authors were not contacted and code repositories were not consulted.

One convention needs stating plainly, because it is the weakest point of the scheme. A threshold, rescaling or selection step never mentioned was coded not performed; a step mentioned without saying which data it was fit on was coded unclear. Silence does not establish that a step was not performed, so the convention can only move studies toward the leakage-free end, and our sheets record one value for both readings. We therefore do not rely on it: Section V reports the strictest alternative, treating every such silence as unclear, and states which conclusions survive it.

C. Rating scheme

Each study received one of four ratings, applied from reported methods alone:

- *Leakage-free*: subject-level partitioning is stated, and connectivity estimation, thresholding, normalization and channel or band selection are each stated to be confined to the training fold or not performed.
- *Leakage-possible*: subject-level partitioning is stated, but at least one of L2–L5 is not clearly described as within-fold.
- *Leakage-likely*: partitioning is at epoch level, or a preprocessing step is explicitly applied to the full dataset before partitioning.
- *Unclear*: the partitioning unit is not stated.

Unclear is a category in its own right rather than missing data, because its prevalence is itself a result for RQ1.

Raters did not assign the rating directly: they coded the partitioning unit and the four surface fields, and a fixed rule implemented in code derived the rating, so both raters' ratings follow the same logic. Where rules overlap, leakage-likely takes precedence: an explicit whole-dataset step makes a study leakage-likely whatever its partitioning unit.

Two conventions were fixed before coding and given to both raters in identical written instructions: a threshold applied to each matrix using only that matrix's own values, and normalization computed within each segment, were coded within-fold, since neither draws on other samples; and batch-normalization layers inside a network were not treated as preprocessing.

D. Rater procedure and analysis plan

Two raters coded every included study independently, without discussion, from identical written instructions. Percentage agreement and unweighted Cohen's κ on the four-level rating were computed before any disagreement was discussed. Disagreements were then resolved with a senior researcher as tiebreaker, and both pre- and post-resolution ratings are reported. A κ below 0.4 was pre-specified as indicating that the scheme is too subjective to use as defined.

The analysis was specified, and its code written, before accuracies were examined. RQ1, RQ2 and RQ4 are descriptive and no confirmatory test is performed for them. For RQ1 we report the proportion of studies in each reporting category, overall and per surface, with Wilson 95% confidence intervals. The 95% interval for κ comes from 10,000 bootstrap resamples of studies.

Four analyses were added after review, once ratings and accuracies were known: the strict re-reading above, per-surface agreement, a bootstrap resampling datasets rather than studies, and an exact test for the year trend. They are post hoc, labeled as such wherever they appear, and changed no coded field, rating or pre-specified result.

RQ3 is the only confirmatory analysis. We report median accuracy and interquartile range per rating stratum and compare the leakage-likely and leakage-free strata with a two-sided Mann–Whitney U test, and all strata with a Kruskal–Wallis test, each accompanied by Cliff's δ with a 10,000-resample bootstrap interval. The smallest effect of interest was fixed in advance at 3 percentage points of median accuracy; a smaller difference is reported as a null, and no subgroup was searched for an effect. Because small datasets may both favor weaker partitioning and yield higher accuracy, accuracy by rating is also reported by sample-size tercile. All resampling used a fixed seed, so every reported value can be regenerated from the coding sheet.

V. Results

A. Corpus characteristics

The audit included 18 studies: 14 from the Web of Science set and 4 from the supplementary search (Fig. 2). Table I lists each. They were published in 2020–2025. The Mumtaz resting-state dataset was used by 8 studies, MODMA by 5, PRED+CT by 2 and a private cohort by 2; 1 used another dataset. The median sample was 61 subjects (range 48–122). Convolutional or recurrent networks were used in 10 studies and graph neural networks in 6; Table I gives each study's connectivity measure. Because several studies analyzed the same public dataset and 3 came from one research group, the studies are not fully independent (Section VII). They fall into only 5 dataset clusters, Mumtaz supplying 8 studies and 6 of the 9 epoch-level ones, so resampling datasets rather than studies widens the interval on the epoch-level proportion below to 0–68% (post hoc). These proportions describe the corpus, not the literature.

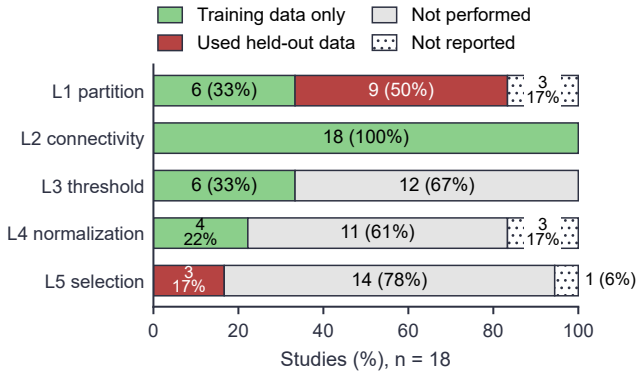


Fig. 3. Reporting completeness by leakage surface ($n = 18$). Segments give the number and percentage of studies in which the step was stated to use training data only, stated to use held-out data, not performed, or not reported; for L1, partitioning by subject and by epoch.

B. Reporting completeness by leakage surface (RQ1)

In 22% of studies the methods did not report enough to determine whether every preprocessing step (L2–L5) was confined to the training fold (95% CI 9–45%), and 50% partitioned at the epoch level (29–71%). Fig. 3 gives the distribution for each surface.

Connectivity was estimated from each segment or subject on its own in every study (L2). Thresholds (L3) and normalization (L4) were either not used or fit per matrix or on training data, except that normalization was not described in 17%. Channel or band selection (L5) drew on all the data in 17% and was unclear in 6%. The steps specific to connectivity (L2–L4) were never reported as fit on the whole dataset; leakage risk arose instead at partitioning and at selection, and the partitioning unit and the normalization step were the ones most often left undescribed. In the secondary analysis (RQ4), reporting completeness fell with publication year (Spearman $\rho = -0.52$, 95% CI -0.82 to -0.02), but an exact test on the dichotomized measure does not support it (odds ratio 10.0, Fisher's $p = 0.11$, post hoc). From 18 studies, confounded by venue and design, this is a hypothesis for a larger corpus, not a finding.

C. Rating distribution and agreement (RQ2)

Before any discussion, the two raters assigned the same rating to 94% of studies, unweighted Cohen's $\kappa = 0.91$ (95% CI 0.66 to 1.00), above the pre-specified threshold of 0.4. On the individual fields they agreed on 17 of 18 studies for the partitioning unit, 18 for connectivity scope, 17 for thresholds, 15 for normalization and 12 for channel or band selection. Because the overall κ is dominated by the most common decisions, we also computed it per surface, post hoc: $\kappa = 0.91$ (95% CI 0.69 to 1.00) for the partitioning unit, 0.87 for thresholds, 0.69 for normalization, and 0.36 (-0.05 to 0.71) for channel or band selection; for connectivity scope every code was identical, so κ is undefined. The surface carrying our main result is thus the most reliably coded and selection the least: its κ falls below the 0.4 value pre-specified for the overall rating, so every L5 figure here should be read as provisional. Disagreement concentrated there because papers compare bands or channels without saying whether the comparison chose the final model; since the rating follows fixed rules, most field-level disagreements left it unchanged.

Studies reporting several protocols were handled as described in Section IV. The tiebreaker resolved the remaining disagreement. After resolution, 3 studies (17%) were rated leakage-free, 2 (11%) leakage-possible, 10 (56%) leakage-likely and 3 (17%) unclear; the 95% intervals were 6–39%, 3–33%, 34–75% and 6–39% respectively.

How much the not-reported convention matters (post hoc). Re-reading every unmentioned threshold, normalization or selection step as unclear rather than not performed—37 coded cells—moves 3 studies from leakage-free to leakage-possible, leaving 0. The leakage-likely count is unchanged at 10, as is the epoch-level proportion, because both rest on what papers state rather than on what they omit. The convention cannot have inflated the central result, then, but it accounts entirely for the leakage-free count: read strictly, no study here can be verified leakage-free from its methods section.

D. Accuracy by rating (RQ3)

The primary comparison used the 17 studies whose recorded accuracy came from a binary depression-versus-control task; one three-class study was excluded from it and enters only the sensitivity analysis. That excluded study is one of the 3 rated leakage-free, which is why 2 and not 3 leakage-free studies appear in this comparison. Median reported accuracy was 98.7% (IQR 97.2–99.3) among the 10 studies rated leakage-likely and 88.9% (the two values were 77.8% and 100.0%) among the 2 rated leakage-free, a difference of 9.8 percentage points (bootstrap 95% CI -2.8 to 21.5; Mann–Whitney $U = 10.0$, p = 1.00; Cliff's $\delta = 0.00$, 95% CI -1.00 to 1.00). Across all 4 strata, a Kruskal–Wallis test gave $H = 3.96$, p = 0.27. The difference in medians exceeds the pre-specified smallest effect of interest of 3 points, but with only 2 leakage-free studies, whose median is the midpoint of their two accuracies, it is not distinguishable from zero: the rank-based effect size is null and both intervals include no difference. This comparison is therefore uninformative. It is not evidence that leakage risk is associated with accuracy, nor evidence that it is not.

Across sample-size terciles (small, medium, large), leakage-likely studies numbered 4/6/0 and leakage-free studies 0/2/1, too few per cell to compare accuracy within terciles.

The pre-specified sensitivity analyses did not change this reading. With one study per research group ($n = 16$) and without the two weakest inclusions ($n = 16$), Cliff's δ was 0.00 and 0.00; and including every task ($n = 18$) gave $\delta = 0.33$; in all three the interval for δ spanned -1 to 1. With each abstract's headline figure, only 1 study was rated leakage-free, too few for any comparison.

VI. A Reporting Checklist

Table II lists eight items a connectivity-based study should state, one per leakage surface, each answerable in a sentence; together they let a reader decide from the paper alone whether test information could have reached training, a question 22% of the audited studies left open for at least one step. For items 5 to 7 what matters is which data a step was fit on, and the epochs per subject (item 2) show how much a subject-identity shortcut could contribute under epoch-level partitioning. The checklist adds no procedure: a study that already partitions by subject and fits every statistic within the training fold satisfies it with a few sentences. Each item should be answered even when the step was not performed, so a reader can tell an omitted step from an undescribed one—a distinction this audit could not draw (Sections IV, V).

General reporting guidance for prediction models and medical-imaging AI—TRIPOD+AI, CLAIM—already asks for the validation scheme and the handling of predictors, and recommendations for EEG decoding ask for subject-level partitioning [25], but none names the statistics specific to a connectivity matrix. Table II replaces none of them; it adds that layer.

VII. Discussion and Limitations

The clearest finding concerns partitioning, not the connectivity pipeline itself: 50% of studies split recordings by epoch and 17% did not state the unit, while no study reported fitting connectivity estimation, thresholds or normalization on the whole dataset. What limited verification was omission. The accuracy comparison could not answer RQ3. That absence of evidence should not be read as evidence that leakage leaves accuracy unchanged, which the neuroimaging studies in Section II show it need not, nor should any individual study's accuracy be read as inflated because of its rating.

This audit rates reporting, not conduct. A study rated unclear or leakage-possible may have run a leakage-free pipeline and not said so; one rated leakage-free may contain an error its text does not reveal. Coding an unmentioned step as not performed overstates how many are leakage-free, which Section V bounds. Our ratings describe what a reader can verify from the paper—the question that matters for how an accuracy should be read—not any author's practice.

TABLE I

CORPUS CHARACTERISTICS, ONE ROW PER INCLUDED STUDY; STUDY GIVES THE REFERENCE NUMBER. RATINGS DESCRIBE WHAT EACH PAPER REPORTS, NOT HOW THE STUDY WAS CONDUCTED.

Study	Dataset	N subj.	Connectivity	Architecture	Partition unit	Acc. (%)	Risk rating from reported methods
[16]	Mumtaz	64	Granger	CNN-LSTM	epoch	99.2	likely
[11]	private	48	coherence	CNN	subject	80.7	likely
[22]	Mumtaz	57	other	CNN	subject	91.1	possible
[12]	Mumtaz	64	PLV	CNN	epoch	97.3	likely
[13]	MODMA	53	PLI	CNN-LSTM	epoch	96.3	likely
[21]	PRED+CT	63	other	GCN	subject	77.8	free
[19]	Mumtaz	55	other	CNN	epoch	99.9	likely
[20]	MODMA	53	other	GCN	epoch	99.7	likely
[34]	MODMA	53	correlation	other	unclear	89.0	unclear
[35]	MODMA	53	correlation	GCN	epoch	98.1	likely
[36]	other	122	correlation	LSTM	unclear	88.6	unclear
[37]	private	67	PLI	CNN	subject	79.6[a]	free
[23]	PRED+CT	120	other	other	unclear	85.3	unclear
[14]	Mumtaz	64	other	CNN	epoch	97.2	likely
[15]	MODMA	53	correlation	GCN	subject	98.3	possible
[38]	Mumtaz	64	other	GCN	epoch	99.3	likely
[17]	Mumtaz	60	Granger	CNN	subject	100.0	free
[18]	Mumtaz	64	PLV	GCN	epoch	99.3	likely

[a] Not a binary depression-vs-control task; excluded from the primary accuracy comparison (Section V-D).

TABLE II

REPORTING CHECKLIST FOR CONNECTIVITY-BASED DEEP LEARNING STUDIES (GEN. = GENERALIZATION)

#	A connectivity study should state	Addresses
1	Whether data were partitioned by subject or by epoch	L1
2	The number of subjects and the number of epochs per subject	L1
3	Whether connectivity was estimated separately within each training fold	L2
4	The window length and overlap used for connectivity estimation	L2
5	How any threshold or edge-count parameter was selected, and from which data	L3
6	Whether feature normalization statistics were computed on training data only	L4
7	Whether channel or band selection used test-set performance	L5
8	Whether any external or cross-dataset evaluation was performed	Gen.

The corpus is not the product of a full systematic review. It began from a previously assembled set of the 100 most-cited Web of Science articles on the topic, whose exact query and search date could not be recovered, and citation-ranked sampling favors older and more visible work. Full-text retrieval stopped once the corpus reached the protocol's minimum of 15 studies, a stopping rule the protocol did not specify; as a result 24 of the 47 screened-in Web of Science reports were not assessed (one because no copy was available to us), and 11 of the 15 supplementary reports were not retrieved. Which studies entered therefore depended partly on retrieval order and access rather than on the criteria alone. The ratings are also not fully independent, several studies sharing a dataset and 3 coming from one research group; Section V reports how much the prevalence figures move when datasets or groups, rather than studies, are resampled.

Both raters worked from identical written instructions and coded independently, but both are authors of this paper, and the rules, though mechanical, required interpretive conventions for thresholds and normalization fixed in advance (Section IV). No rater outside the author group coded any study and no blinded re-coding was performed; the per-surface agreement in Section V stands in its place, and shows the weakness plainly—partitioning is coded reliably, channel and band selection is not. AI-drafted screening, approved by an author (Acknowledgment), may also have missed studies a second human screener would have caught.

Availability. The coding sheet, both raters' codes, the evidence quote for each study, the tiebreak record and all analysis and figure code are at https://github.com/kunalchevuri/eeg-connectivity-leakage-audit. Those quotes, with the rules in Section IV, let a reader re-rate the corpus.

VIII. CONCLUSION

We audited 18 studies applying deep learning to EEG connectivity for depression detection, rating leakage risk from reported methods alone. 50% partitioned by epoch and 22% did not report enough to tell whether preprocessing was confined to the training fold; no study reported fitting connectivity estimation, thresholds or normalization on the whole dataset, and with only 2 leakage-free studies the accuracy comparison was uninformative. Read strictly, treating an unmentioned step as undescribed rather than absent, 0 studies can be shown leakage-free from their text. A pipeline that cannot be reconstructed from its description cannot be checked for leakage; Table II asks for the eight statements that make the check possible.

CONTRIBUTIONS AND ACKNOWLEDGMENT

K.C. designed the audit and wrote the protocol, the analysis code and the manuscript. K.C. and L.I. were the two raters,

each coding every study independently from identical instructions and without discussion. M.A. supervised the work and, not being a rater, acted as tiebreaker. All authors approved the final manuscript.

Generative AI was used as follows; the authors reviewed all of it and take responsibility for the content. Claude (Anthropic; Claude Code and claude.ai, with Opus 5, Opus 5.5 and Sonnet 5) drafted the screening decisions, which an author approved; ran the PubMed, OpenAlex and Semantic Scholar searches; drafted the rater instructions; extracted descriptive metadata, which an author checked against each paper (for seven studies the AI instead re-checked the values against the full text); wrote the analysis, table and figure code to a specification fixed before accuracies were examined; drafted or revised text in every section, which the authors edited; and cross-checked cited claims against sources. Perplexity was used to find candidate references, run Google Scholar searches and draft reference-check notes. An author opened and checked every reference. No AI tool coded a leakage field or rated a study.

REFERENCES

[1] A. J. Ferrari *et al.*, "Burden of Depressive Disorders by Country, Sex, Age, and Year: Findings from the Global Burden of Disease Study 2010," *PLoS Medicine*, vol. 10, no. 11, p. e1001547, 2013.

[2] H. Lin *et al.*, "Resting-State Electroencephalogram Depression Diagnosis Based on Traditional Machine Learning and Deep Learning: A Comparative Analysis," *Sensors*, vol. 24, no. 21, p. 6815, 2024.

[3] D. A. Regier *et al.*, "DSM-5 Field Trials in the United States and Canada, Part II: Test-Retest Reliability of Selected Categorical Diagnoses," *American Journal of Psychiatry*, vol. 170, no. 1, pp. 59–70, 2013.

[4] L. Simmatis *et al.*, "Technical and clinical considerations for electroencephalography-based biomarkers for major depressive disorder," *npj Mental Health Research*, vol. 2, no. 1, p. 18, 2023.

[5] R. H. Kaiser *et al.*, "Large-Scale Network Dysfunction in Major Depressive Disorder," *JAMA Psychiatry*, vol. 72, no. 6, p. 603, 2015.

[6] G. Kaur *et al.*, "Artificial intelligence driven neuropsychiatry: a systematic review of electroencephalography-based computational techniques for major depressive disorder prediction," *Neuroscience*, vol. 581, pp. 179–207, 2025.

[7] M. Rosenblatt *et al.*, "Data leakage inflates prediction performance in connectome-based machine learning models," *Nature Communications*, vol. 15, no. 1, p. 1829, 2024.

[8] E. Yagis *et al.*, "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," *Scientific Reports*, vol. 11, no. 1, p. 22544, 2021.

[9] H.-T. Lee *et al.*, "Risk of data leakage in estimating the diagnostic performance of a deep-learning-based computer-aided system for psychiatric disorders," *Scientific Reports*, vol. 13, no. 1, p. 16633, 2023.

[10] G. R. Luna-Guevara *et al.*, "Systematic review of machine learning and deep learning models for EEG-based detection of depression," *Journal of Psychiatric Research*, vol. 199, pp. 113–121, 2026.

[11] X. Li *et al.*, "A Deep Learning Approach for Mild Depression Recognition Based on Functional Connectivity Using Electroencephalography," *Frontiers in Neuroscience*, vol. 14, p. 192, 2020.

[12] W. Dang *et al.*, "Multilayer brain network combined with deep convolutional neural network for detecting major depressive disorder," *Nonlinear Dynamics*, vol. 102, no. 2, pp. 667–677, 2020.

[13] B. Zhang *et al.*, "Spatial–Temporal EEG Fusion Based on Neural Network for Major Depressive Disorder Detection," *Interdisciplinary Sciences: Computational Life Sciences*, vol. 15, no. 4, pp. 542–559, 2023.

[14] H.-S. Yu *et al.*, "Attention-mechanism–based network characteristic analysis for major depressive disorder detection," *Europhysics Letters*, vol. 139, no. 1, p. 11004, 2022.

[15] W. Liu *et al.*, "Graph-based EEG approach for depression prediction: integrating time-frequency complexity and spatial topology," *Frontiers in Neuroscience*, vol. 18, p. 1367212, 2024.

[16] A. Saeedi *et al.*, "Major depressive disorder diagnosis based on effective connectivity in EEG signals: a convolutional neural network and long short-term memory approach," *Cognitive Neurodynamics*, vol. 15, no. 2, pp. 239–252, 2021.

[17] D. M. Khan *et al.*, "Automated Diagnosis of Major Depressive Disorder Using Brain Effective Connectivity and 3D Convolutional Neural Network," *IEEE Access*, vol. 9, pp. 8835–8846, 2021.

[18] X. Sun *et al.*, "A Novel Complex Network-Based Graph Convolutional Network in Major Depressive Disorder Detection," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–8, 2022.

[19] L. Li *et al.*, "An EEG-based marker of functional connectivity: detection of major depressive disorder," *Cognitive Neurodynamics*, vol. 18, no. 4, pp. 1671–1687, 2024.

[20] X. Sun *et al.*, "Multi-Granularity Graph Convolution Network for Major Depressive Disorder Recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 559–569, 2024.

[21] G. Luo *et al.*, "Exploring Adaptive Graph Topologies and Temporal Graph Networks for EEG-Based Depression Detection," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 3947–3957, 2023.

[22] M. Xia *et al.*, "An End-to-End Deep Learning Model for EEG-Based Major Depressive Disorder Classification," *IEEE Access*, vol. 11, pp. 41 337–41 347, 2023.

[23] Y. Guo *et al.*, "Integrating Cortical Source Reconstruction and Adversarial Learning for EEG Classification," *Sensors*, vol. 25, no. 16, p. 4989, 2025.

[24] L. Zhang *et al.*, "Machine learning approaches for classifying major depressive disorder using biological and neuropsychological markers: A meta-analysis," *Neuroscience & Biobehavioral Reviews*, vol. 174, p. 106201, 2025.

[25] G. Brookshire *et al.*, "Data leakage in deep learning studies of translational EEG," *Frontiers in Neuroscience*, vol. 18, p. 1373515, 2024.

[26] F. S. Racz and G. Csukly, "Information leakage and performance overestimation in EEG-based schizophrenia detection: evidence from literature and empirical analyses," *Translational Psychiatry*, 2026, in press, doi: 10.1038/s41398-026-04315-9.

[27] J. Wen *et al.*, "Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation," *Medical Image Analysis*, vol. 63, p. 101694, 2020.

[28] V. M. Young *et al.*, "Data Leakage in Deep Learning for Alzheimer's Disease Diagnosis: A Scoping Review of Methodological Rigor and Performance Inflation," *Diagnostics*, vol. 15, no. 18, p. 2348, 2025.

[29] P. L. Nunez *et al.*, "EEG coherency," *Electroencephalography and Clinical Neurophysiology*, vol. 103, no. 5, pp. 499–515, 1997.

[30] J.-P. Lachaux *et al.*, "Measuring phase synchrony in brain signals," *Human Brain Mapping*, vol. 8, no. 4, pp. 194–208, 1999.

[31] C. J. Stam *et al.*, "Phase lag index: Assessment of functional connectivity from multi channel EEG and MEG with diminished bias from common sources," *Human Brain Mapping*, vol. 28, no. 11, pp. 1178–1193, 2007.

[32] Y. Di *et al.*, "The Time-Robustness Analysis of Individual Identification Based on Resting-State EEG," *Frontiers in Human Neuroscience*, vol. 15, p. 672946, 2021.

[33] A. Sahuc, "Evaluation Traps in EEG Disease Classification: Identity Leakage, Lucky Folds, and Objective Mismatch Replicated Across AD, FTD, MDD, and SCZ," Zenodo preprint, 2026.

[34] X. Jia *et al.*, "Multimodal depression detection based on an attention graph convolution and transformer," *Mathematical Biosciences and Engineering*, vol. 22, no. 3, pp. 652–676, 2025.

[35] Y. Zeng *et al.*, "Adaptive node feature extraction in graph-based neural networks for brain diseases diagnosis using self-supervised learning," *NeuroImage*, vol. 297, p. 120750, 2024.

[36] B. Prakash *et al.*, "A Novel Approach to Learning Models on EEG Data Using Graph Theory Features—A Comparative Study," *Big Data and Cognitive Computing*, vol. 5, no. 3, p. 39, 2021.

[37] Y. Li *et al.*, "EEG assessment of brain dysfunction for patients with chronic primary pain and depression under auditory oddball task," *Frontiers in Neuroscience*, vol. 17, p. 1133834, 2023.

[38] Y. Chen *et al.*, "A Local-Global Graph Convolutional Network for Depression Recognition using EEG Signals," *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 7, 2023.